M3403 WEEK 6 2020/21

LAPLACE TRANSFORM

Let $f(t)$ be defined on $[0,\infty)$. The integral

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = F(s)$$

is the laplace transform of $f(t)$ provided that it converges.

Recall that $\int_0^\infty e^{-st} f(t)dt = \lim_{a\to\infty} \int_0^a e^{-st} f(t)\,dt$

Examples

Determine the laplace transform of

① $f(t) = 1$

② $f(t) = e^{at}$

③ $f(t) = \sin(at)$

④ $f(t) = \begin{cases} 0, & 0<t<2 \\ t, & 2>t \end{cases}$

sol^n

① $\mathcal{L}\{1\} = \int_0^\infty e^{-st}\cdot 1\,dt$

$= \lim_{a\to\infty} \int_0^a e^{-st}\,dt$

$= \lim_{a\to\infty} \left[-\frac{1}{s} e^{-st}\right]_0^a$

$= \lim_{a\to\infty} \left[-\frac{1}{s} e^{-as} + \frac{1}{s}\right]$

$= \frac{1}{s}$

(2) $\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st} e^{at}\, dt$

$= \lim_{N\to\infty} \int_0^N e^{(a-s)t}\, dt$

$= \lim_{N\to\infty} \left[\frac{1}{a-s} e^{(a-s)t} \right]_0^N$

$= \lim_{N\to\infty} \left[\frac{1}{a-s} e^{(a-s)N} - \frac{1}{a-s} \right]$

$= \frac{1}{s-a}$ if $a < s$

(3) $\mathcal{L}\{\sin(at)\} = \int_0^\infty e^{-st} \sin(at)\, dt$

$= \lim_{N\to\infty} \int_0^N e^{-st} \sin(at)\, dt$

$= \lim_{N\to\infty} \left[\frac{1}{1+a^2/s^2} \left(-\frac{1}{s} e^{-st} \sin(at) - \frac{a}{s^2} e^{-st} \cos(at) \right) \right]_0^N$

$= \lim_{N\to\infty} \left[\frac{1}{1+a^2/s^2} \left(-\frac{1}{s} e^{-sN} \sin(aN) - \frac{a}{s^2} e^{-sN} \cos(aN) + \frac{a}{s^2} \right) \right]$

$= \frac{a}{s^2(1+a^2/s^2)}$

$= \frac{a}{s^2 + a^2}$

④. $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$

$$= \int_0^2 e^{-st} \cdot 0\, dt + \int_2^\infty e^{-st} t\, dt$$

$$= 0 + \lim_{N\to\infty} \int_2^N e^{-st} t\, dt$$

$$= \lim_{N\to\infty} \left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_2^N$$

$$= \lim_{N\to\infty} \left[-\frac{N}{s} e^{-sN} - \frac{1}{s^2} e^{-sN} + \frac{2}{s} e^{-2s} + \frac{1}{s^2} e^{-2s} \right]$$

$$= \left(\frac{1}{s^2} + \frac{2}{s} \right) e^{-2s}$$

Linearity Of Laplace Trasform

Suppose the laplace transforms of f_1 and f_2 exist for $s > a$, then

① $\mathcal{L}\{f_1 + f_2\} = \mathcal{L}\{f_1\} + \mathcal{L}\{f_2\}$; for $s > a$

② $\mathcal{L}\{c f_1\} = c\,\mathcal{L}\{f_1\}$; for $s > a$

Exponential Order

A function $f(t)$ is said to be of exponential order c if there exit positive constants M and T such that

$$|f(t)| \leq M e^{ct} \quad \forall\ t \geq T$$

Piecewise Continuity

A function $f(t)$ is piecewise continuous on $[a, b]$ if it is continuous at every point in $[a, b]$, except for a finite number of points at which $f(t)$ has a jump discontinuity.

Existence Of Laplace Transform

If $f(t)$ is piecewise continuous on $[0, \infty)$ and of exponential order a, then $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > a$.

Unless stated otherwise, the entries in the following table can be use to determine Laplace transforms of some basic functions

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$
t^n, $n = 1, 2, \ldots$	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(at)$	$\frac{a}{s^2+a^2}$
$\cos(at)$	$\frac{s}{s^2+a^2}$
$e^{at}t^n$, $n = 1, 2, \ldots$	$\frac{n!}{(s-a)^{n+1}}$

$e^{at}\sin(kt)$	$\frac{k}{(s-a)^2+k^2}$
$e^{at}\cos(kt)$	$\frac{s-a}{(s-a)^2+k^2}$
$\sinh(kt)$	$\frac{k}{s^2-k^2}$
$\cosh(kt)$	$\frac{s}{s^2-k^2}$

PROPERTIES OF LAPLACE TRANSFORM

① If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > \alpha$, then
$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$ for $s > \alpha + a$
This is known as the first shift theorem.

② If $f(t)$ is continuous on $[0,\infty)$, and $f'(t)$ is piecewise continuous on $[0,\infty)$, with both of exponential order α, then

$$\mathcal{L}\{f'(t)\} = s\,\mathcal{L}\{f(t)\} - f(0), \quad \text{for } s > \alpha$$

③. If $f(t), f'(t), \ldots, f^{(n-1)}(t)$ are continuous on $[0,\infty)$ and $f^{(n)}(t)$ is piecewise continuous on $[0,\infty)$, with all these functions of exponential order α, then for $s > \alpha$

$$\mathcal{L}\{f^{(n)}(t)\} = s^n\mathcal{L}\{f(t)\} - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

④ If $f(t)$ is piecewise continuous on $[0,\infty)$ and of exponential order α, then for $s > \alpha$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}, \quad \text{where } F(s) = \mathcal{L}\{f(t)\}$$

Examples

Find the Laplace transform of

① $f(t) = 2t^2e^{-t} - t + \cos 4t$

② $f(t) = e^{-t}t\sin 2t$

③ $f(t) = \sin^2 t$

sol^n

① $\mathcal{L}\{2t^2e^{-t} - t + \cos 4t\}$

$= 2\mathcal{L}\{t^2e^{-t}\} - \mathcal{L}\{t\} + \mathcal{L}\{\cos(4t)\}$

$= 2\,\frac{2!}{(s-(-1))^{(2+1)}} - \frac{1!}{s^{(1+1)}} + \frac{s}{s^2+4^2}$

$= \frac{4}{(s+2)^3} - \frac{1}{s^2} + \frac{s}{s^2+16}$

② $\mathcal{L}\{e^{-t}t\sin(2t)\}$

$= (-1)\frac{d}{ds}\mathcal{L}\{e^{-t}\sin(2t)\}$

$= -\frac{d}{ds}\left[\frac{2}{(s+1)^2+4}\right]$

$= -\frac{d}{ds}\left[\frac{2}{s^2+2s+5}\right]$

$= -2\left[\frac{-(2s+2)}{(s^2+2s+5)^2}\right]$

$= \frac{4(s+1)}{(s^2+2s+5)^2}$

(8). $\mathcal{L}\{\sin^2 t\} = \mathcal{L}\left\{\frac{1-\cos(2t)}{2}\right\}$

$= \frac{1}{2}\mathcal{L}\{1\} - \frac{1}{2}\mathcal{L}\{\cos(2t)\}$

$= \frac{1}{2s} - \frac{s}{2(s^2+4)}$

Inverse Trasform

Given a function $F(s)$, if there exists a function $f(t)$ that is continuous on $[0,\infty)$, and

$$\mathcal{L}\{f(t)\} = F(s)$$

then $f(t)$ is the inverse Laplace transform of $F(s)$ i.e $f(t) = \mathcal{L}^{-1}\{F(s)\}$

Examples

Determine the inverse transform of the given function

1. $F(s) = \frac{1}{s^3}$

2. $F(s) = \frac{6}{(s-1)^4}$

3. $F(s) = \frac{s+1}{s^2+2s+10}$

4. $F(s) = \frac{5s^2+34s+53}{(s+3)^2(s+1)}$

5. $F(s) = \ln\left(\frac{s+2}{s-5}\right)$

Basically, we refer to the basic transforms given previously, in a table, and laplace transform properties.

Sol 4.

① $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{1}{2!}\mathcal{L}^{-1}\left\{\frac{2!}{s^3}\right\}$

$= \frac{1}{2}t^2$

② $\mathcal{L}^{-1}\left\{\frac{6}{(s-1)^4}\right\} = \frac{6}{3!}\mathcal{L}^{-1}\left\{\frac{3!}{(s-1)^4}\right\}$

$= e^t t^3$

③ $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2s+10}\right\} = \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+9}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{s-(-1)}{(s-(-1))^2+3^2}\right\}$

$= e^{-t}\cos(3t)$

④ $\mathcal{L}^{-1}\left\{\frac{5s^2+3+5s+53}{(s+3)^2(s+1)}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{-1}{s+3} + \frac{2}{(s+3)^2} + \frac{6}{s+1}\right\}$ using partial fractions

$= \mathcal{L}^{-1}\left\{-\left[\frac{1}{s-(-3)}\right] + 2\left[\frac{1}{(s-(-3))^2}\right] + 6\left[\frac{1}{s-(-1)}\right]\right\}$

$= -e^{-3t} + 2te^{-3t} + 6e^{-t}$

(5) $\mathcal{L}^{-1}\left\{\ln\left(\frac{s+2}{s-5}\right)\right\}$

From property (4) of Laplace transform

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}\mathcal{L}\{f(t)\}$$

$$\Rightarrow (-t)^n f(t) = \mathcal{L}^{-1}\left\{\frac{d^n}{ds^n}\mathcal{L}\{f(t)\}\right\}$$

$$\Rightarrow f(t) = \left(\frac{-1}{t}\right)^n \mathcal{L}^{-1}\left\{\frac{d^n}{ds^n}\mathcal{L}\{f(t)\}\right\}$$

Now, $\mathcal{L}^{-1}\left\{\ln\left(\frac{s+2}{s-5}\right)\right\} = \frac{-1}{t}\mathcal{L}^{-1}\left\{\frac{d}{ds}\left[\ln\left(\frac{s+2}{s-5}\right)\right]\right\}$

$$= \frac{-1}{t}\mathcal{L}^{-1}\left\{\frac{-7}{(s+2)(s-5)}\right\}$$

$$= \frac{1}{t}\mathcal{L}^{-1}\left\{\frac{-1}{s+2} + \frac{1}{s-5}\right\}$$

$$= \frac{1}{t}\left[-e^{-2t} + e^{5t}\right] \quad //$$